

Fluids and Lubricants Specifications

MTU Fluids and Lubricants Specifications for Series 1600 PowerPack®

A001065/00E



Power. Passion. Partnership.

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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1 Preface

1.1 General information

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:



This symbol indicates instructions, tasks and operations that must be followed to avoid hazards to persons as well as damage to or destruction of material.

Note:

A note provides special instructions that must be observed when performing a task.

Fluids and lubricants

The service life, operational reliability and function of the drive systems are largely dependent on the fluids and lubricants employed. The correct selection and treatment of these fluids and lubricants are therefore extremely important. This publication specifies which fluids and lubricants are to be used.

Test standard	Designation
DIN	Federal German Standards Institute
EN	European Standards
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
IP	Institute of Petroleum
DVGW	German Gas and Water Industry Association

Table 1: Test standards for fluids and lubricants

Applicability of this publication

The Fluids and Lubricants Specifications will be amended or supplemented as necessary. Before using them, make sure you have the latest version. The latest version is also available at:

www.mtu-online.com/mtu/technical-info/fluids-and-lubricants-specifications/

If you have further queries, please contact your MTU representative.

Warranty

Use of the approved fluids and lubricants, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the fluids and lubricants is responsible for the worldwide standard quality of the named products.



Fluids and lubricants for propulsion plants may be considered hazardous substances. Certain regulations must be obeyed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, statutory regulations and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. MTU accepts no responsibility whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

Preservation

All information on preservation, re-preservation and de-preservation including the approved preservatives is available in the MTU Preservation and Re-preservation Specifications (publication number A001070/...). The latest version is also available at:

www.mtu-online.com/mtu/technical-info/preservation-specification/

2 Engine Oils

2.1 Requirements and oil change intervals

	Dispose of used fluids and lubricants in accordance with local regulations. Used oil must never be disposed of via the fuel tank!
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Requirements of the engine oils for MTU approval

The MTU conditions for engine oil approval are specified in the MTU Factory Standard MTL 5044, which can be ordered under this reference number.

Manufacturers of engine oils are notified in writing if their product is approved.

Approved engine oils for Series 1600 engines are divided into the following MTU Quality Categories:

- Oil category 3: Highest quality / Multi-grade oils
- Oil category 3.1: Multi-grade oils with a low ash-forming additive content (low SAPS oils)

Low SAPS oils are oils with a low sulfur and phosphor content and an ash-forming additive content of $\leq 1\%$. They are only approved if the sulfur content in the fuel does not exceed 500 mg/kg. When using diesel particle filters, it is advisable to use these oils to avoid fast coating of the filter with ash particles.

Selection of a suitable engine oil is based on fuel quality, projected oil drain interval and on-site climatic conditions. At present there is no international industrial standard which alone takes into account all these criteria.

	The use of engine oils not approved by MTU can mean that statutory emission limits can no longer be observed. This can be a punishable offense.
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Oil change interval

	The oil change interval is 1,000 operating hours or max. 1 year under the condition that engine oils of oil category 3 (→ Page 21) and 3.1 (→ Page 24) and approved fuels (→ Page 17) are used. If fuels which have not been approved are used, shorter oil change intervals are to be expected. Prior to using unapproved fuels, contact MTU Friedrichshafen GmbH to determine the applicable oil change intervals.
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	Mixing different engine oils is strictly prohibited!
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Changing to another oil grade can be done together with an oil change. The remaining oil quantity in the engine oil system is not critical in this regard.

2.2 Viscosity grades

Selection of the viscosity grade is based primarily on the ambient temperature at which the engine is to be started and operated. Figure (→ Figure 1) contains guideline values for the temperature limits of the individual viscosity grades.

If the prevailing temperature is too low, the engine oil must be preheated.

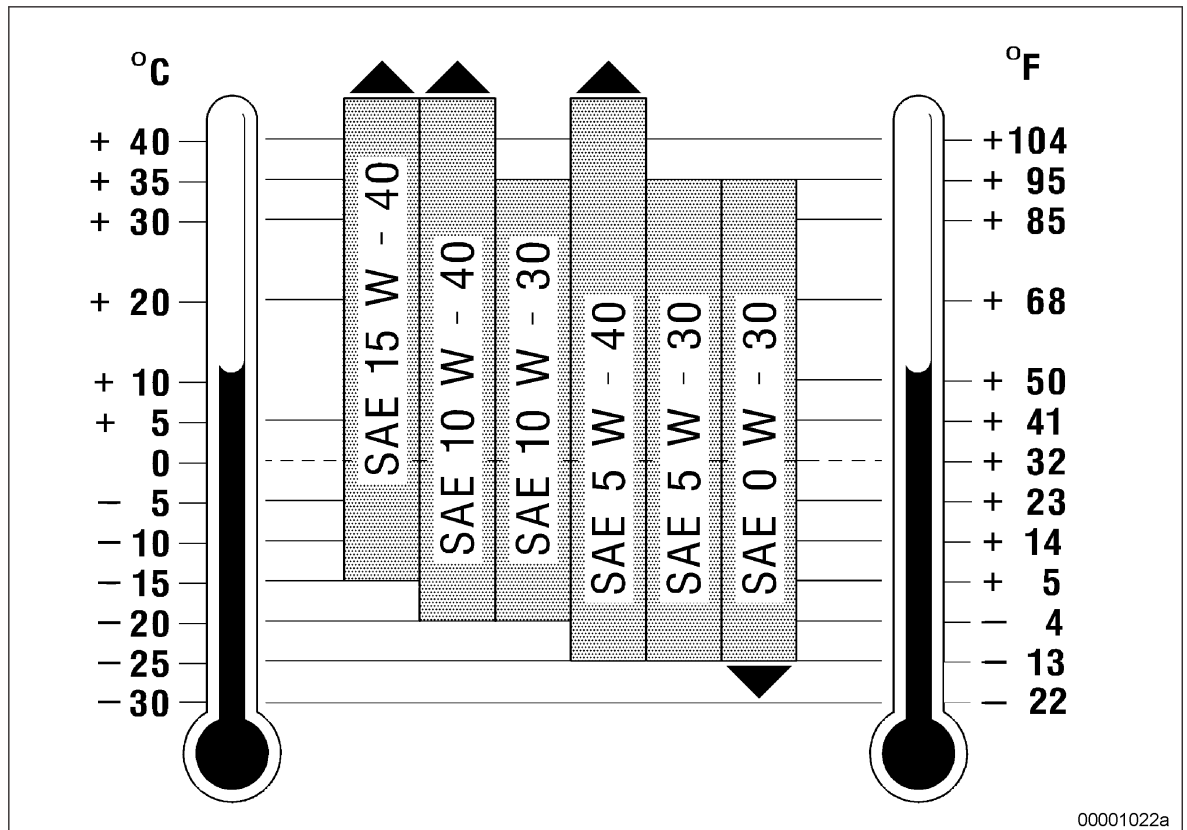


Figure 1: Viscosity grades

3 Transmission oils

3.1 Transmission oils for rail vehicles with ZF transmission

(Abstract from the ZF List of Lubricants TE-ML16, Edition 01.10.2008)

The ZF Lists of Lubricants are updated every three months on 01.01., 01.04., 01.07. and 01.10.. Before using them, make sure you have the latest version. The latest version is also available at:

www.zf.com/corporate/de/products/spare_parts/technical_information/lubricants/lists_of_lubricants.html/TE-ML16

Product groups automatic transmissions for rail vehicles	Lubricant classes for service fills ⁽¹⁾ transmissions with-out/with ZF-Intarder
ASRail 12 AS 2303. 12 AS 2703. 12 AS 3103. 16 AS2 603	16K / 16P
EcoLife (up to 105 °C)	16Q

⁽¹⁾ = Approves commercial products (→ Page 26), oil change intervals and low temperature limits (listed below).

⁽²⁾ = Particularly recommended: The fully synthetic ATF ZF-Ecofluid A PLUS was developed specifically for use in Ecomat transmissions. This combination of a synthetic base oil with a specially balanced additive package delivers superlative oxidation stability and consistent friction characteristics. The viscosity level is ideally suited to this transmission and this factor, combined with resistance to scuffing and pitting, assures the unit of extremely valuable protection and a correspondingly long service life for its bearings and gears. Another positive feature of ZF-Ecofluid A PLUS is its flat viscosity characteristics curve which makes it particularly well suited to operation in cold as well as in hot climatic zones .

At greasing points pay attention to the manual!

	Additives of any kind added later to the oil change the oil in a manner that is unpredictable, and they are therefore not permitted. No liability of any kind will be accepted by ZF for any damage resulting from the use of such additives.
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Oil and filter change intervals for EcoLife transmissions for rail vehicles:

Lubricant classes ⁽¹⁾	Oil and filter change interval [km / years] ^(2,3)
16Q	180,000 km / every 3 years

⁽¹⁾ = Pay attention to approved trade products and lubricant classes.

⁽²⁾ = Oil change required, depending on what occurs first.

⁽³⁾ = After consultation with the product support department of ZF Friedrichshafen AG, Special Drive Technology, and after an oil analysis has been made (after agreed mileages), longer oil change intervals can be applied to some reference transmissions. The procedure for taking oil samples is described in the respective Service Information.

Application areas of lubricants

The following illustration (→ Figure 2) shows application areas of the various SAE classes in relation to the ambient temperatures to be expected.

The oils have a bottom limit of max. dynamic viscosity (Brookfield) of 150,000 mPas, which roughly corresponds to the viscosity limit at low temperatures.

The upper limit is determined by the load in the transmission and the appearing temperature level during operation. It can be assumed that high ambient temperatures will also result in higher oil sump temperatures. For detailed information on the low temperature behavior of the specific product see at the safety data sheet of the supplier.

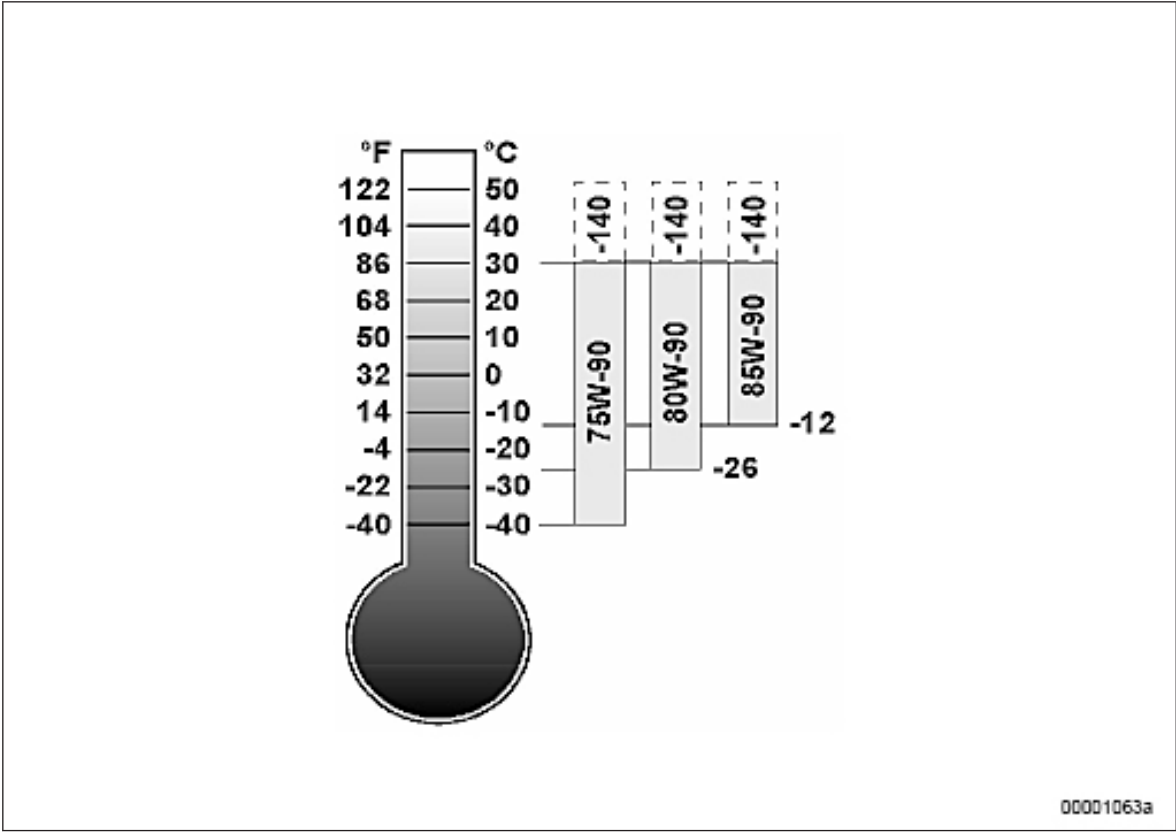


Figure 2: Range of use for hydraulic oil

The user must observe the low temperature limits!

Lubricant classes	Viscosity grades	Use at oil sump temperature as over
16K / 16L / 16M / 16N / 16P / 16Q	75W-80 / 75W-85 / 75W-90 / 75W-110 / 75W-140 / ATF	- 40 °C

3.2 Power transmission oils for Voith turbo transmission T211 + KB 190

Excerpt from Voith documentation of power transmission oils for Voith turbo transmissions

Voith publications are continuously updated. Before using them, make sure you have the latest version. The latest version is also available at:

www.voith.com/brochures/2255

Oil and filter change intervals for Voith turbo transmissions T 211 re4 + KB190

Oil and filter change interval based on running hours ⁽¹⁾	Oil and filter change intervals based on mileage (km)
5,000	300,000

⁽¹⁾ = Running hours are accumulated operating hours at speeds of more than 1 km/h.

Use at low temperatures

The approved transmission oils allow cold starts at temperatures down to -20 °C.

Special measures must be taken if temperatures are lower.

Oil filtration

Ensure oil filtration to purity class 15/11 as per ISO 4406 when filling the transmission with oil. For appropriate filter units please contact Voith Turbo.

The maximum quantity of foreign particles in 100 ml oil for this purity class is:

- Particles >4µm: 130 000 (purity class 17)
- Particles >6µm: 32 000 (purity class 15)
- Particles >14µm: 2 000 (purity class 11)

Approved power transmission oils (→ Page 27).